

Compositions of Trigonometric Functions and Their Inverses, and Back Substitution

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1 The problem

A trigonometric function is not injective, so it has no inverse on its natural domain. What we call $\arcsin$, $\arccos$, and the others are inverses of *restrictions*: we pick an interval on which the function is injective and invert it there. Every subtlety in this essay comes from that restriction.

There are three kinds of composition to understand.

1. $f(f^{-1}(x))$, for example $\sin(\arcsin x)$. This is the easy direction.
2. $f^{-1}(f(x))$, for example $\arcsin(\sin x)$. This equals x only on the restricted interval; elsewhere it folds x back into that interval.
3. $g(f^{-1}(x))$ with $g \neq f$, for example $\tan(\operatorname{arcsec} x)$. This is what back substitution in an integral asks for, and it is where sign errors occur.

The single principle that resolves all of them is this: *an inverse trigonometric function returns an angle in a specific interval, and every sign question is answered by looking at that interval.* Square roots never carry sign information; $\sqrt{u^2} = |u|$, always.

2 Principal ranges

Definition 2.1. The six inverse trigonometric functions used in this essay are defined as follows. For each, $\theta = f^{-1}(x)$ is the unique θ in the stated range with $f(\theta) = x$.

function	domain in x	range in θ
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$
$\operatorname{arccot} x$	$\mathbb{R}$	$(0, \pi)$
$\operatorname{arcsec} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi/2) \cup (\pi/2, \pi]$
$\operatorname{arccsc} x$	$(-\infty, -1] \cup [1, \infty)$	$[-\pi/2, 0) \cup (0, \pi/2]$

The ranges are chosen to satisfy two requirements: the original function must be injective on the range, and the range must be as close to 0 as possible while covering every value. Notice the pattern by quadrant. $\arcsin$, $\arctan$, and arccsc take values in quadrants I and IV. $\arccos$, arccot , and arcsec take values in quadrants I and II. In each case, positive x gives a first-quadrant angle, and negative x gives an angle in the other quadrant of the pair.

Remark 2.2. These conventions are not universal. Some texts define arccot with range $(-\pi/2, 0) \cup (0, \pi/2]$, so that $\operatorname{arccot} x = \arctan(1/x)$ for $x \neq 0$. Some calculus texts define arcsec with range $[0, \pi/2) \cup [\pi, 3\pi/2)$, chosen so that $\tan(\operatorname{arcsec} x) \geq 0$ always. Every formula in Section 5 depends on the convention. Check which one your source uses before comparing answers.

With the conventions above, arcsec and arccsc reduce to the more familiar functions:

$$\operatorname{arcsec} x = \arccos \frac{1}{x}, \quad \operatorname{arccsc} x = \arcsin \frac{1}{x}, \quad |x| \geq 1.$$

Indeed, $\sec \theta = x$ if and only if $\cos \theta = 1/x$, and both sides take values in the same interval. This identity is often the quickest route through a computation.

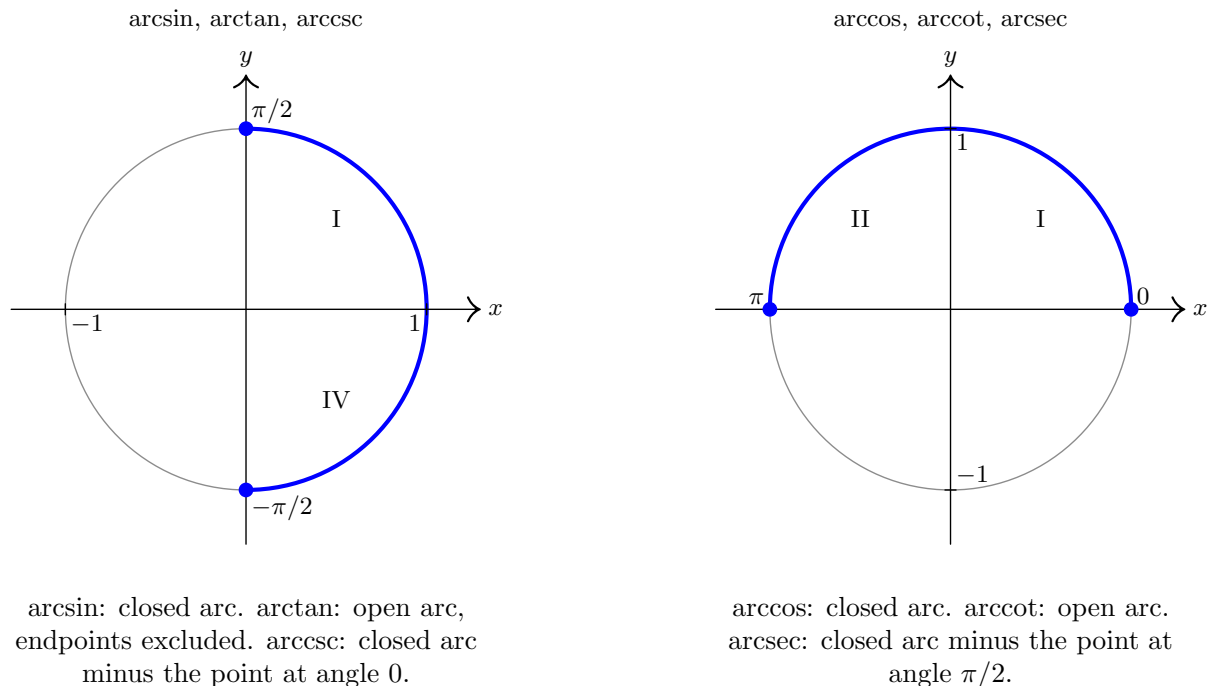


Figure 1: Principal ranges of the inverse trigonometric functions, shown on the unit circle.

3 The easy direction: $f(f^{-1}(x))$

Proposition 3.1. *For each of the six functions, $f(f^{-1}(x)) = x$ for every x in the domain of f^{-1} , and is undefined otherwise.*

Proof. By definition $f^{-1}(x)$ is an angle θ with $f(\theta) = x$. □

There is nothing to compute, but the domain matters: $\sin(\arcsin 2)$ is not 2; it is undefined. Likewise $\sec(\operatorname{arcsec} \frac{1}{2})$ is undefined. $\tan(\arctan x) = x$ and $\cot(\operatorname{arccot} x) = x$ hold for every real x .

4 The folding direction: $f^{-1}(f(x))$

Here x is an arbitrary angle and f^{-1} must return an angle in its range. If x is already in the range, we get x back. Otherwise we get the unique angle in the range that has the same value of f as x . Graphically, $f^{-1}(f(x))$ is a piecewise linear function of slope ± 1 .

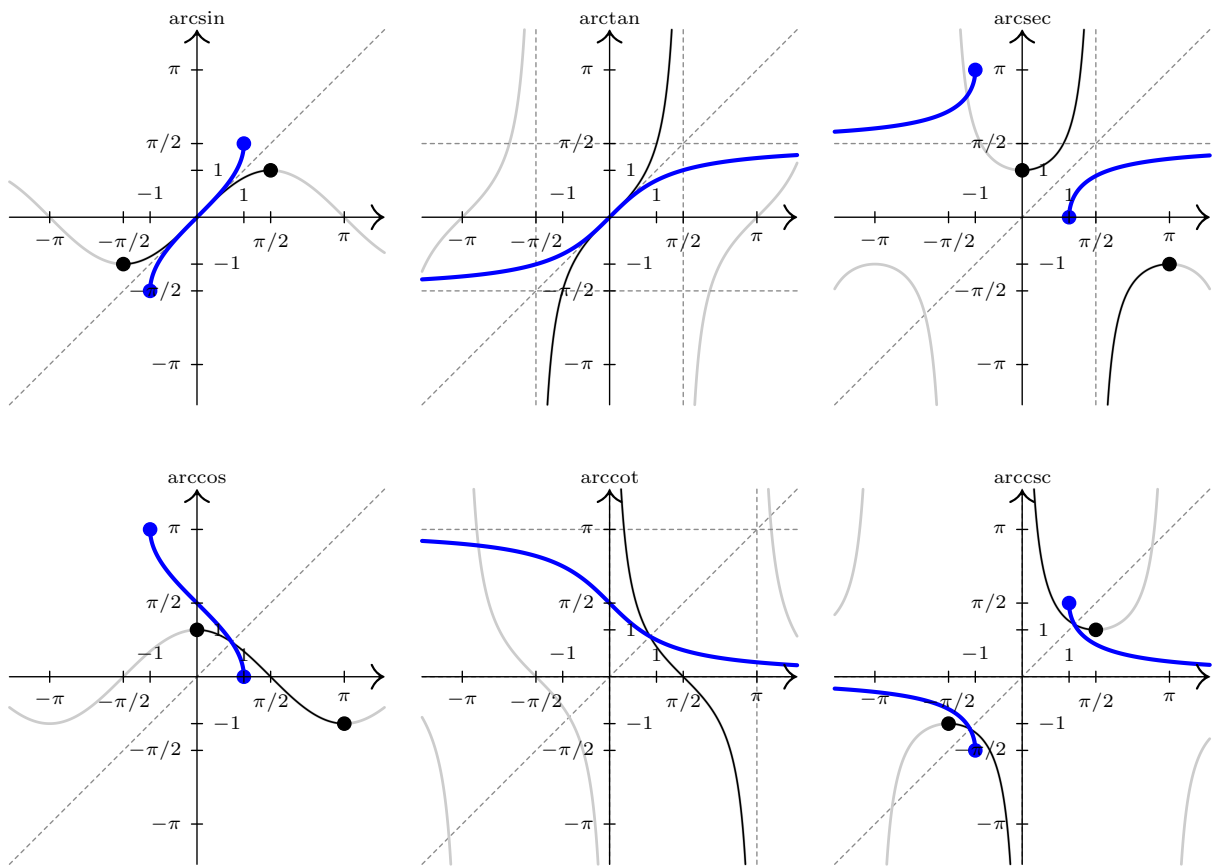


Figure 2: Restricted trigonometric functions and their inverses. Each inverse is the reflection of the restricted function in the line $y = x$.

Proposition 4.1. For every integer k ,

$$\begin{aligned}\arcsin(\sin x) &= (-1)^k(x - k\pi), & x &\in [k\pi - \frac{\pi}{2}, k\pi + \frac{\pi}{2}], \\ \arccos(\cos x) &= |x - 2k\pi|, & x &\in [(2k-1)\pi, (2k+1)\pi], \\ \arctan(\tan x) &= x - k\pi, & x &\in (k\pi - \frac{\pi}{2}, k\pi + \frac{\pi}{2}).\end{aligned}$$

Proof. In each case, check two things: the right-hand side lies in the range of the inverse function, and f of the right-hand side equals $f(x)$.

For $\arcsin$: if $x \in [k\pi - \pi/2, k\pi + \pi/2]$ then $x - k\pi \in [-\pi/2, \pi/2]$, and so is its negative. Also $\sin((-1)^k(x - k\pi)) = (-1)^k \sin(x - k\pi) = (-1)^k(-1)^k \sin x = \sin x$, using $\sin(x - k\pi) = (-1)^k \sin x$ and the oddness of $\sin$.

For $\arccos$: if $x \in [(2k-1)\pi, (2k+1)\pi]$ then $x - 2k\pi \in [-\pi, \pi]$, so $|x - 2k\pi| \in [0, \pi]$. Since $\cos$ is even and 2π -periodic, $\cos|x - 2k\pi| = \cos(x - 2k\pi) = \cos x$.

For $\arctan$: $x - k\pi \in (-\pi/2, \pi/2)$, and $\tan$ has period π . □

Example 4.2. $\arcsin(\sin 3)$. Since $3 \in [\pi/2, 3\pi/2]$, take $k = 1$: the result is $-(3 - \pi) = \pi - 3 \approx 0.14$. Check: $\pi - 3$ lies in $[-\pi/2, \pi/2]$, and $\sin(\pi - 3) = \sin 3$.

Example 4.3. $\arccos(\cos(-1)) = 1$, because $\cos$ is even and $1 \in [0, \pi]$. Note that $\arcsin(\sin(-1)) = -1$. The two folds differ because the ranges differ.

Example 4.4. $\arctan(\tan 2) = 2 - \pi$, since $2 \in (\pi/2, 3\pi/2)$.

5 Cross compositions: $g(f^{-1}(x))$

This is the case that matters for integration. We know one trigonometric value of an angle, $f(\theta) = x$, and the interval containing θ . We want a different trigonometric value $g(\theta)$ as an algebraic expression in x .

5.1 The method

1. Set $\theta = f^{-1}(x)$. Write down what you know: $f(\theta) = x$ and the interval containing θ .
2. Use a Pythagorean identity,

$$\sin^2 \theta + \cos^2 \theta = 1, \quad 1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta,$$

to find $g(\theta)^2$, or the square of some function from which $g(\theta)$ follows without a further square root.

3. Determine the sign of $g(\theta)$ on the interval. If it is constant, take the root with that sign. If it is not constant, the sign depends on which part of the range θ lies in, which in turn depends on the sign of x ; express it through x .
4. Simplify with $\sqrt{x^2} = |x|$, never $\sqrt{x^2} = x$.
5. Check the domain: exclude x for which $g(\theta)$ is undefined.

Step 3 is the whole content of the method. It is cheap to carry out, and skipping it is what produces wrong answers.

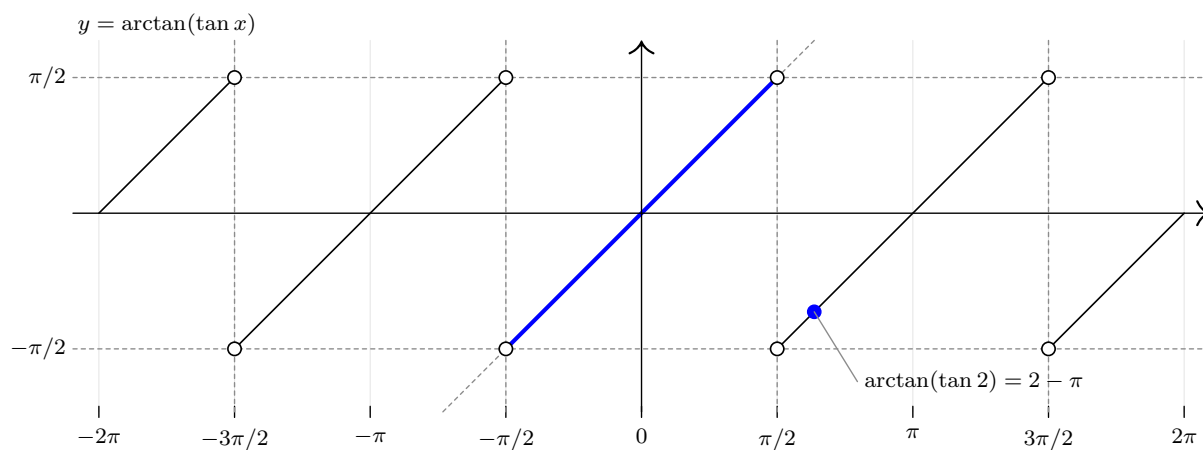
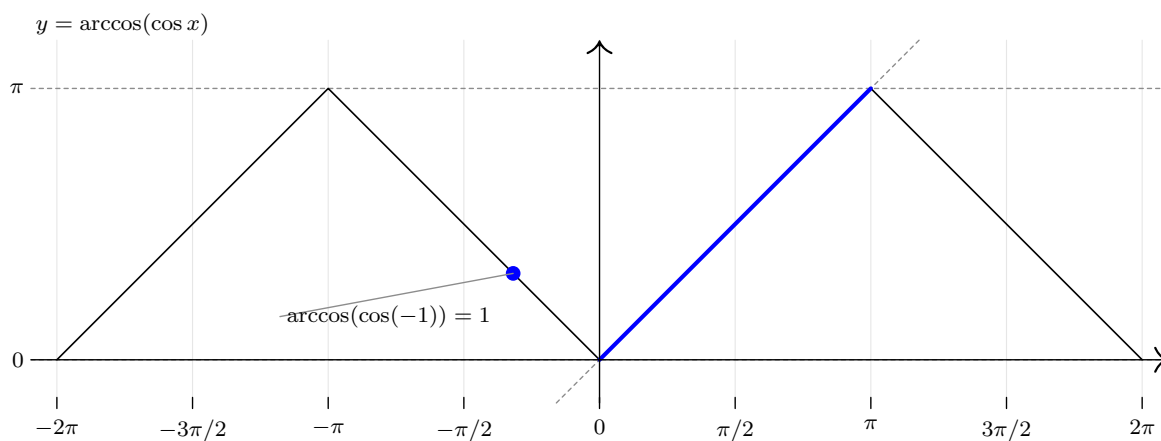
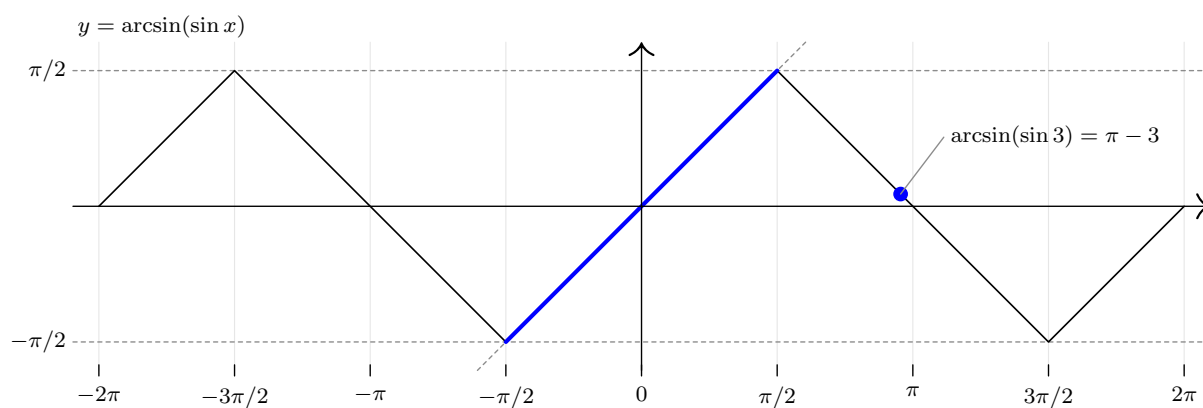


Figure 3: The folding compositions on $[-2\pi, 2\pi]$. Each graph is the identity on the principal range, shown thick, and is folded back into the range elsewhere.

5.2 A worked case where the sign is constant

Example 5.1. Consider $\csc(\operatorname{arcsec} x)$. Let $\theta = \operatorname{arcsec} x$. Then $\cos \theta = 1/x$ and $\theta \in [0, \pi/2) \cup (\pi/2, \pi]$, so

$$\sin^2 \theta = 1 - \frac{1}{x^2} = \frac{x^2 - 1}{x^2}.$$

On $[0, \pi]$ we have $\sin \theta \geq 0$, on both parts of the range. Therefore

$$\sin \theta = \frac{\sqrt{x^2 - 1}}{\sqrt{x^2}} = \frac{\sqrt{x^2 - 1}}{|x|}, \quad \csc(\operatorname{arcsec} x) = \frac{|x|}{\sqrt{x^2 - 1}}, \quad |x| > 1.$$

The points $x = \pm 1$ are excluded because there $\theta \in \{0, \pi\}$ and $\sin \theta = 0$.

The absolute value appears because $\sin \theta$ keeps its sign across the range while $x = \sec \theta$ changes sign. A formula written in terms of x alone must undo that sign change.

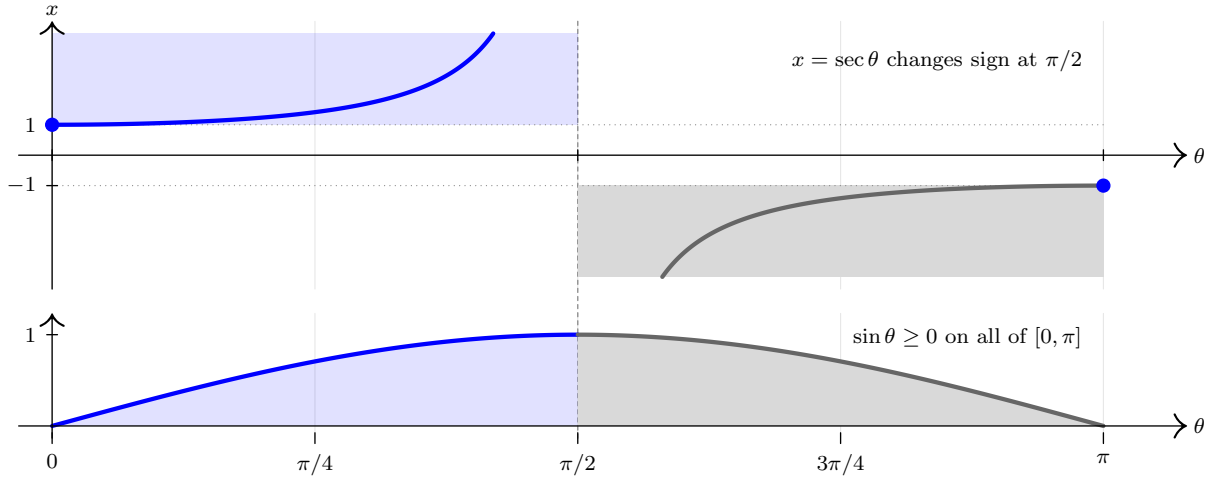


Figure 4: On the range of arcsec , $x = \sec \theta$ changes sign at $\pi/2$ (top) while $\sin \theta$ stays nonnegative (bottom). The two halves of the range are shown in the same colours in both panels.

5.3 A worked case where the sign changes

Example 5.2. $\tan(\operatorname{arcsec} x)$. With θ as before, $\tan^2 \theta = \sec^2 \theta - 1 = x^2 - 1$. Now $\tan \theta \geq 0$ on $[0, \pi/2)$, which is the case $x \geq 1$, and $\tan \theta \leq 0$ on $(\pi/2, \pi]$, which is the case $x \leq -1$. So

$$\tan(\operatorname{arcsec} x) = \begin{cases} \sqrt{x^2 - 1}, & x \geq 1, \\ -\sqrt{x^2 - 1}, & x \leq -1, \end{cases}$$

that is, $\tan(\operatorname{arcsec} x) = \operatorname{sgn}(x)\sqrt{x^2 - 1}$ (see Figure 5, where the sign of $\tan \theta$ matches the sign of x on each part of the range). Alternatively, $\tan \theta = \sin \theta / \cos \theta = \frac{\sqrt{x^2 - 1}}{|x|} \cdot x$, which gives the same result, since $x/|x| = \operatorname{sgn} x$. Computing a sign-stable function first and deriving the others from it is often the cleanest route.

5.4 A uniform tool: the signed reference point

The familiar reference-triangle method draws a right triangle with acute angle θ and reads off side lengths. It is correct only when $\theta \in (0, \pi/2)$, that is, only for positive x in most cases. For the other half of the range it silently gives the wrong sign.

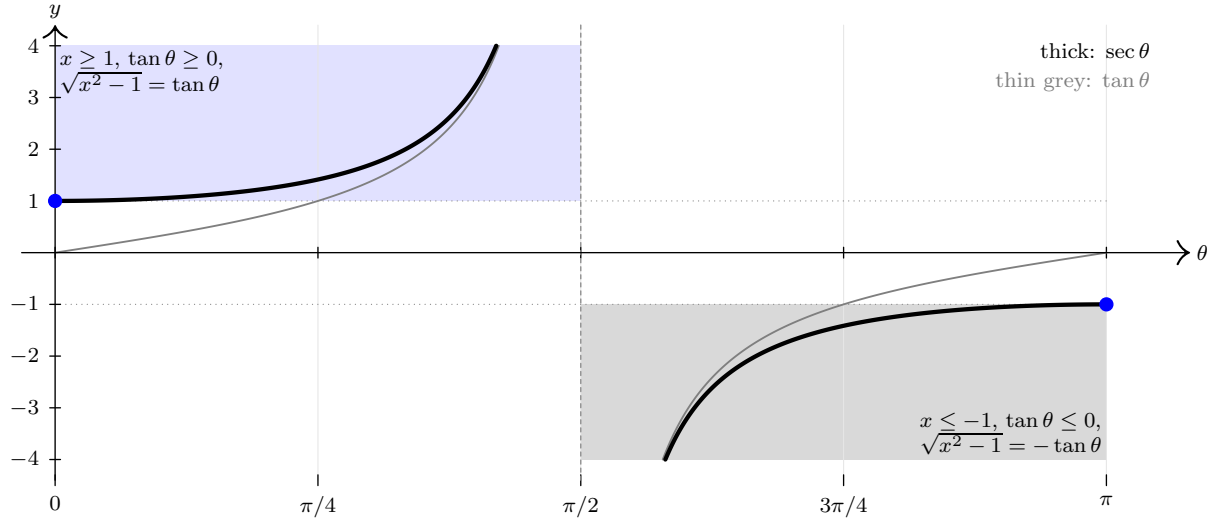


Figure 5: For $\theta = \operatorname{arcsec} x$, the sign of $\tan \theta$ equals the sign of x : both are nonnegative on $[0, \pi/2)$ and both are nonpositive on $(\pi/2, \pi]$.

The correct generalisation replaces the triangle with a point (X, Y) on the terminal side of θ , at distance $r = \sqrt{X^2 + Y^2} > 0$ from the origin. Then

$$\begin{aligned} \sin \theta &= \frac{Y}{r}, & \cos \theta &= \frac{X}{r}, & \tan \theta &= \frac{Y}{X}, \\ \csc \theta &= \frac{r}{Y}, & \sec \theta &= \frac{r}{X}, & \cot \theta &= \frac{X}{Y}. \end{aligned}$$

Here X and Y are signed coordinates, and the quadrant of θ dictates their signs. To evaluate $g(f^{-1}(x))$:

1. Choose X, Y, r consistent with $f(\theta) = x$, keeping $r > 0$.
2. Fix the sign of the remaining coordinate from the range of f^{-1} .
3. Read off $g(\theta)$.

Example 5.3. $\theta = \operatorname{arcsec} x$, so $r/X = x$. Since $r > 0$, X has the sign of x ; take $X = \operatorname{sgn} x$ and $r = |x|$. Then $Y^2 = r^2 - X^2 = x^2 - 1$, and since $\theta \in [0, \pi]$, $Y \geq 0$, so $Y = \sqrt{x^2 - 1}$. Every value follows at once:

$$\begin{aligned} \sin \theta &= \frac{\sqrt{x^2 - 1}}{|x|}, & \tan \theta &= \frac{\sqrt{x^2 - 1}}{\operatorname{sgn} x} = \operatorname{sgn}(x) \sqrt{x^2 - 1}, \\ \csc \theta &= \frac{|x|}{\sqrt{x^2 - 1}}, & \cot \theta &= \frac{\operatorname{sgn} x}{\sqrt{x^2 - 1}}. \end{aligned}$$

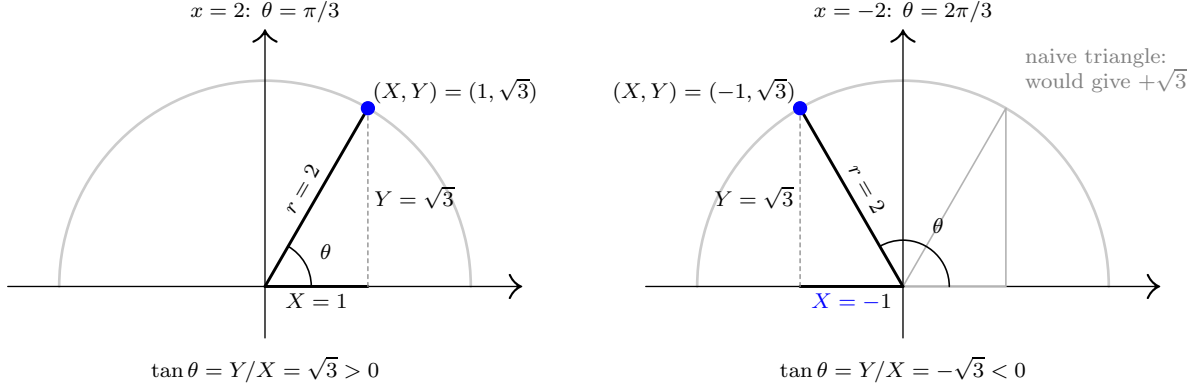


Figure 6: The reference triangle versus the signed reference point for $\theta = \arcsin x$. For $x = -2$ the angle is in quadrant II, and the triangle picture, which assumes an acute angle, gives the wrong sign for $\tan \theta$.

5.5 Table of cross compositions

The following table collects the common cases under the conventions of Section 2. The last column states where the sign of the result comes from.

expression	value	valid for	sign source
$\sin(\arccos x)$	$\sqrt{1-x^2}$	$ x \leq 1$	$\sin \geq 0, [0, \pi]$
$\cos(\arcsin x)$	$\sqrt{1-x^2}$	$ x \leq 1$	$\cos \geq 0, [-\frac{\pi}{2}, \frac{\pi}{2}]$
$\tan(\arcsin x)$	$\frac{x}{\sqrt{1-x^2}}$	$ x < 1$	$\operatorname{sgn} \sin \theta = \operatorname{sgn} x$
$\tan(\arccos x)$	$\frac{x}{\sqrt{1-x^2}}$	$0 < x \leq 1$	$\operatorname{sgn} \cos \theta = \operatorname{sgn} x$
$\sin(\arctan x)$	$\frac{x}{\sqrt{1+x^2}}$	$\mathbb{R}$	$\cos > 0, (-\frac{\pi}{2}, \frac{\pi}{2})$
$\cos(\arctan x)$	$\frac{1}{\sqrt{1+x^2}}$	$\mathbb{R}$	same
$\sin(\operatorname{arccot} x)$	$\frac{x}{\sqrt{1+x^2}}$	$\mathbb{R}$	$\sin > 0, (0, \pi)$
$\cos(\operatorname{arccot} x)$	$\frac{1}{\sqrt{1+x^2}}$	$\mathbb{R}$	same
$\sin(\operatorname{arcsec} x)$	$\frac{\sqrt{x^2-1}}{ x }$	$ x \geq 1$	$\sin \geq 0, [0, \pi]$
$\tan(\operatorname{arcsec} x)$	$\operatorname{sgn}(x)\sqrt{x^2-1}$	$ x \geq 1$	branch-dependent
$\csc(\operatorname{arcsec} x)$	$\frac{ x }{\sqrt{x^2-1}}$	$ x > 1$	$\sin \geq 0, [0, \pi]$
$\cos(\operatorname{arccsc} x)$	$\frac{\sqrt{x^2-1}}{ x }$	$ x \geq 1$	$\cos \geq 0, [-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cot(\operatorname{arccsc} x)$	$\operatorname{sgn}(x)\sqrt{x^2-1}$	$ x \geq 1$	branch-dependent

A pattern is visible. In every row, the result has the form $\pm\sqrt{\cdot}$ times a power of x or $|x|$. Whether x or $|x|$ appears is decided by whether the target function changes sign across the range of the inverse function. Memorising the table is unnecessary; rederiving any row takes three lines once the range is known.

6 Back substitution in integrals

In a trigonometric substitution we set $x = af(\theta)$, integrate in θ , and obtain an answer containing trigonometric functions of θ . To return to x we must evaluate expressions such as $\sin \theta$ or $\tan \theta$ with $\theta = f^{-1}(x/a)$. That is exactly a cross composition, and everything in Section 5 applies.

For the substitution to be legitimate, $x = af(\theta)$ must be invertible on the interval of integration, so θ really is $f^{-1}(x/a)$ there. This is what fixes the interval for θ ; it is not a matter of choice once the inverse function's convention is set.

6.1 Substitutions without sign trouble

For $x = a \sin \theta$ with $\theta \in [-\pi/2, \pi/2]$, and for $x = a \tan \theta$ with $\theta \in (-\pi/2, \pi/2)$, the radical becomes $a \cos \theta$ or $a \sec \theta$, both nonnegative on the whole interval. One interval covers the entire domain in x , and no case split arises.

Example 6.1. $\int \sqrt{1-x^2} dx$. Put $x = \sin \theta$, $\theta \in [-\pi/2, \pi/2]$, so $\sqrt{1-x^2} = \cos \theta$ and $dx = \cos \theta d\theta$:

$$\int \cos^2 \theta d\theta = \frac{\theta + \sin \theta \cos \theta}{2} + C.$$

Back substitution: $\theta = \arcsin x$, $\sin \theta = x$, and $\cos \theta = \cos(\arcsin x) = \sqrt{1-x^2}$ from the table. Hence

$$\int \sqrt{1-x^2} dx = \frac{\arcsin x + x\sqrt{1-x^2}}{2} + C, \quad |x| \leq 1.$$

Note that the double-angle form $\frac{1}{4} \sin 2\theta$ must be expanded to $\frac{1}{2} \sin \theta \cos \theta$ before back substituting. Writing $\sin(2 \arcsin x)$ is correct but not algebraic.

Example 6.2. $\int \frac{dx}{x^2 \sqrt{x^2+4}}$. Put $x = 2 \tan \theta$, $\theta \in (-\pi/2, \pi/2)$, $\theta \neq 0$. Then $\sqrt{x^2+4} = 2 \sec \theta$ and $dx = 2 \sec^2 \theta d\theta$:

$$\int \frac{2 \sec^2 \theta}{4 \tan^2 \theta \cdot 2 \sec \theta} d\theta = \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta = -\frac{1}{4 \sin \theta} + C.$$

Back substitution: $\sin \theta = \sin(\arctan(x/2)) = \frac{x/2}{\sqrt{1+x^2/4}} = \frac{x}{\sqrt{x^2+4}}$. Hence

$$\int \frac{dx}{x^2 \sqrt{x^2+4}} = -\frac{\sqrt{x^2+4}}{4x} + C.$$

The formula for $\sin(\arctan u)$ carries the sign of u on its own, so the result is correct for negative x without any case analysis.

6.2 The secant substitution

For $\sqrt{x^2-a^2}$ the domain $|x| \geq a$ has two components, and no single interval for θ covers both while keeping $\tan \theta$ of one sign. With $x = a \sec \theta$:

$$\sqrt{x^2-a^2} = a|\tan \theta| = \begin{cases} a \tan \theta, & x \geq a, \theta \in [0, \pi/2), \\ -a \tan \theta, & x \leq -a, \theta \in (\pi/2, \pi]. \end{cases}$$

Work each branch separately, then combine.

Example 6.3. $\int \frac{\sqrt{x^2-1}}{x} dx$. Put $x = \sec \theta$, $dx = \sec \theta \tan \theta d\theta$.

Branch $x \geq 1$. $\sqrt{x^2-1} = \tan \theta$, and

$$\int \frac{\tan \theta}{\sec \theta} \sec \theta \tan \theta d\theta = \int \tan^2 \theta d\theta = \tan \theta - \theta + C.$$

Back substitution: $\tan \theta = \sqrt{x^2-1}$, giving $\sqrt{x^2-1} - \operatorname{arcsec} x + C$.

Branch $x \leq -1$. $\sqrt{x^2-1} = -\tan \theta$, so the integrand becomes $-\tan^2 \theta$ and the antiderivative is $-\tan \theta + \theta + C$. Back substitution: here $\tan \theta = -\sqrt{x^2-1}$, so $-\tan \theta = \sqrt{x^2-1}$, giving $\sqrt{x^2-1} + \operatorname{arcsec} x + C$.

Check by differentiation, using $\frac{d}{dx} \operatorname{arcsec} x = \frac{1}{|x|\sqrt{x^2-1}}$. For $x < -1$, $|x| = -x$, and

$$\frac{d}{dx} \left(\sqrt{x^2-1} + \operatorname{arcsec} x \right) = \frac{x}{\sqrt{x^2-1}} - \frac{1}{x\sqrt{x^2-1}} = \frac{x^2-1}{x\sqrt{x^2-1}} = \frac{\sqrt{x^2-1}}{x}.$$

Combining. Since $\operatorname{arcsec}(-u) = \pi - \operatorname{arcsec} u$, on $x \leq -1$ we have $\operatorname{arcsec} x = \pi - \operatorname{arcsec} |x|$, so the second branch equals $\sqrt{x^2-1} - \operatorname{arcsec} |x|$ up to a constant. A single formula valid on both components is

$$\int \frac{\sqrt{x^2-1}}{x} dx = \sqrt{x^2-1} - \operatorname{arcsec} |x| + C.$$

Since $\operatorname{arcsec} |x| = \arctan \sqrt{x^2-1}$ for $|x| \geq 1$, this can also be written $\sqrt{x^2-1} - \arctan \sqrt{x^2-1} + C$, which avoids arcsec altogether. (Strictly, the constant may differ on the two components, as with any antiderivative on a disconnected domain.)

Remark 6.4. Had we used the acute reference triangle on the second branch, we would have written $\tan \theta = \sqrt{x^2-1}$ and obtained $-\sqrt{x^2-1} + \operatorname{arcsec} x$, whose derivative is the negative of the integrand for $x < -1$. The error is invisible unless one differentiates on the negative branch.

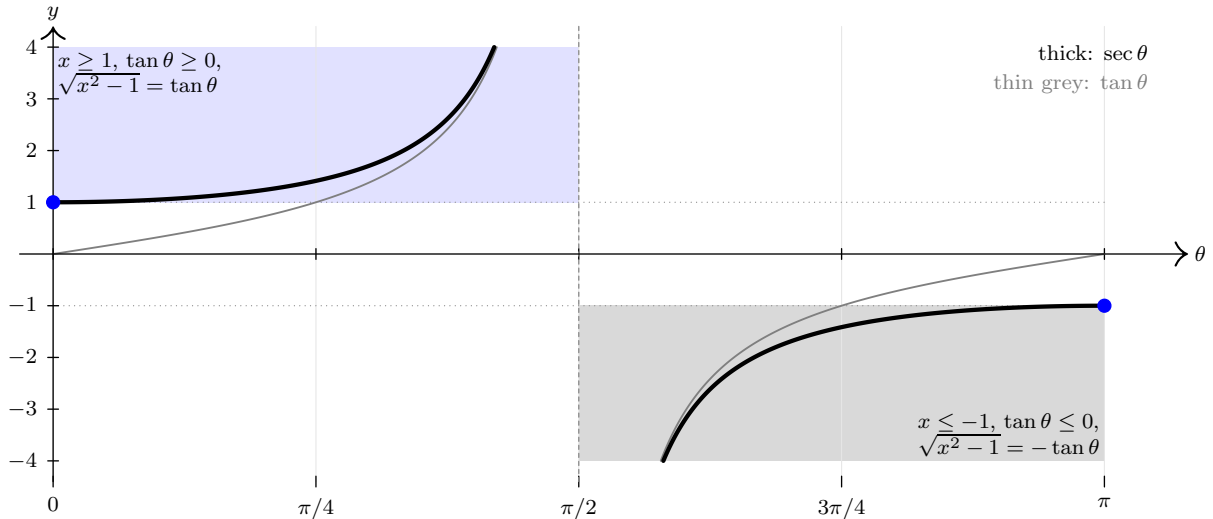


Figure 7: The substitution $x = \sec \theta$ maps the two parts of the θ -range onto the two components of the x -domain. The sign of $\tan \theta$, and hence the relation between $\sqrt{x^2-1}$ and $\tan \theta$, differs on the two parts.

6.3 Definite integrals: change the limits instead

For a definite integral one can avoid back substitution entirely by converting the limits to θ . The branch issue does not disappear, but it becomes explicit: the limits tell you which part of the range you are in.

Example 6.5. $\int_{-2}^{-\sqrt{2}} \frac{dx}{x\sqrt{x^2-1}}$. The whole interval lies in $x \leq -1$, so with $x = \sec \theta$ we have $\theta \in (\pi/2, \pi]$ and $\sqrt{x^2-1} = -\tan \theta$. The integrand becomes

$$\frac{\sec \theta \tan \theta}{\sec \theta \cdot (-\tan \theta)} d\theta = -d\theta.$$

The limits are $\operatorname{arcsec}(-2) = 2\pi/3$ and $\operatorname{arcsec}(-\sqrt{2}) = 3\pi/4$. Hence

$$\int_{-2}^{-\sqrt{2}} \frac{dx}{x\sqrt{x^2-1}} = -\left(\frac{3\pi}{4} - \frac{2\pi}{3}\right) = -\frac{\pi}{12}.$$

Sanity check: on $[-2, -\sqrt{2}]$ the integrand is negative (negative numerator sign from x , positive radical) and the interval is traversed left to right, so the integral must be negative. Ignoring the sign of $\tan \theta$ would have produced $+\pi/12$.

7 Summary

The inverse trigonometric functions are inverses of restrictions, and every composition is governed by the restricted interval. $f(f^{-1}(x)) = x$ on the domain of f^{-1} . $f^{-1}(f(x))$ folds x into the range of f^{-1} and equals x only there. A cross composition $g(f^{-1}(x))$ is computed with a Pythagorean identity, and its sign is read from the range of f^{-1} , not from a square root and not from an acute triangle. When g keeps its sign across the range while x does not, an absolute value appears; when g changes sign with x , the result carries the sign of x .

Back substitution is the same computation. For sine and tangent substitutions the relevant signs are constant and nothing goes wrong. For secant and cosecant substitutions the x -domain splits in two, the θ -range splits accordingly, and each branch must be handled on its own before the answers are combined. The reliable check is to differentiate the result on each branch separately.

8 Self-check questions

1. Why must a trigonometric function be restricted before it has an inverse? For which x is $f(f^{-1}(x)) = x$ defined?
2. Explain why $\arcsin(\sin x)$ need not equal x . Find $\arcsin(\sin(5\pi/6))$ and $\arccos(\cos(5\pi/3))$, checking that each answer lies in the appropriate principal range.
3. What happens at $x = \pi/2$ in $\arctan(\tan x)$? Why does this composition have jumps, whereas $\arcsin(\sin x)$ does not?
4. What is $\sqrt{x^2}$ for arbitrary real x ? Under what condition may you replace it by x ? Explain how this matters when a Pythagorean identity is used to find a trigonometric function of an inverse angle.
5. Derive $\tan(\arccos x)$ from an identity and the range of $\arccos$. State its domain, including any points where the composition is undefined.

6. Derive $\cos(\operatorname{arccsc} x)$ for both signs of x . Explain where the absolute value in the denominator comes from, and check the endpoints of the domain.
7. Compare the signs of $\cos(\operatorname{arccsc} x)$ and $\cot(\operatorname{arccsc} x)$ for $x > 1$ and $x < -1$. Derive the latter expression and explain when $|x|$ or $\operatorname{sgn}(x)$ enters each formula.
8. Let $\theta = \operatorname{arcsec} x$. Choose a signed reference point for θ when $x = -2$, and use it to find $\sin \theta$ and $\tan \theta$. Which sign would an acute reference triangle get wrong?
9. A student proposes $\sin(\operatorname{arcsec} x) = \sqrt{x^2 - 1}/x$ and $\tan(\operatorname{arcsec} x) = \sqrt{x^2 - 1}$ for $|x| \geq 1$. Test both at $x = -2$, correct any errors, and explain from the principal range why one correction uses $|x|$ and the other uses $\operatorname{sgn}(x)$.
10. For $a > 0$ and $x = a \sec \theta$ on the principal range of arcsec , express $\sqrt{x^2 - a^2}$ in terms of θ on each branch. Why is $a \tan \theta$ insufficient on the whole domain?
11. Evaluate $\int_{-2}^{-\sqrt{2}} \frac{\sqrt{x^2 - 1}}{x} dx$ using $x = \sec \theta$. Convert the limits first, and check the sign of your result against the original integrand.
12. The essay gives one antiderivative of $\sqrt{x^2 - 1}/x$ for $|x| > 1$. Why may the constants of integration on $(-\infty, -1)$ and $(1, \infty)$ differ? Verify the formula by differentiating on both intervals.